RIPHAH INTERNATIONAL UNIVERSITY

Data Mining - Semester Project Report on the

Prescriptive Academic Intelligence System

Submitted in partial fulfillment of course requirements for
Data Mining – Semester Project

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Abstract

This project develops a hybrid predictive–prescriptive analytics system using academic performance data. Logistic Regression outperformed Random Forest and XGBoost with an accuracy of **99.92%**, highlighting the linear nature of the target distribution. Beyond prediction, the project introduces a **Prescriptive Analytics Framework** using coefficient analysis and counterfactual simulations. The analysis identifies **x7** as the most influential feature, with targeted improvements demonstrating measurable performance category shifts. The proposed system offers highly accurate prediction and actionable recommendations for educational intervention.

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Chapter 1

Introduction

1.1 Background

Academic performance prediction is a pivotal field in educational data mining (EDM), offering institutions the opportunity to intervene early and improve student outcomes. Historically, studies relied on demographic and attendance data. This project differentiates itself by focusing on fine-grained, quantitative component scores (**x1** to **x7**) to derive behavioral features, enabling a deeper understanding of performance drivers.

1.2 Problem Statement

1. Develop accurate models to predict both the continuous score (*total*) and the categorical performance level (*remarks*).
2. Identify actionable interventions that meaningfully improve student outcomes.

1.3 Gap and Solution (Interventional Focus)

- **The Gap:** Traditional predictive models focus only on *correlation* (identifying factors associated with success) and lack a mechanism to prescribe a concrete course of action. This creates a deployment gap between prediction accuracy and practical utility.
- **The Solution:** This project implements a **Prescriptive Analytics Framework** based on model interpretability (Logistic Regression coefficients) and validated through **Counterfactual Simulation**. The solution provides **Interventional Feature Attribution**, defining the precise $\mathbf{x}i$ that, if improved by a specific amount, yields the largest positive outcome shift.

1.4 Objectives

- To perform comprehensive data preparation, including scaling and engineering critical behavioral features (Consistency Score, Bottleneck Score).
- To implement and compare the performance of at least three hybrid algorithms (Logistic Regression, Random Forest, XGBoost) for the dual tasks of classification and regression.
- To select the best-performing model empirically and utilize its inherent interpretability to identify the most significant factors influencing academic success.

1.5 Scope and Significance

The study uses numerical academic component scores (**x1** to **x7**) and focuses on developing a high-accuracy system ready for integration into an early intervention microservice. The primary significance is the transition from predictive to **prescriptive intelligence**.

Chapter 2

Literature Review

2.1 Performance Prediction in EDM

Traditional Educational Data Mining (EDM) literature widely uses classification and regression algorithms to forecast student outcomes (e.g., pass/fail). However, most models, even highly accurate ones like complex deep learning models, are often criticized for their ”black box” nature and inability to provide direct, ethical, and actionable advice.

2.2 Algorithms and Advances

2.2.1 Classification Methods: Logistic Regression, Random Forest, and XGBoost were chosen to span linear, ensemble, and gradient boosting approaches.

2.2.1 Regression Methods: Linear Regression is utilized due to the additive nature of the underlying data structure.

2.2.1 Prescriptive Analytics: This study directly addresses the need for model transparency and actionability, positioning the work at the forefront of interpretable machine learning application in education.

Chapter 3

Methodology

This project follows the **CRISP-DM** framework.

3.1 Workflow Diagram

The following diagram illustrates the iterative and phased approach followed throughout the project lifecycle.

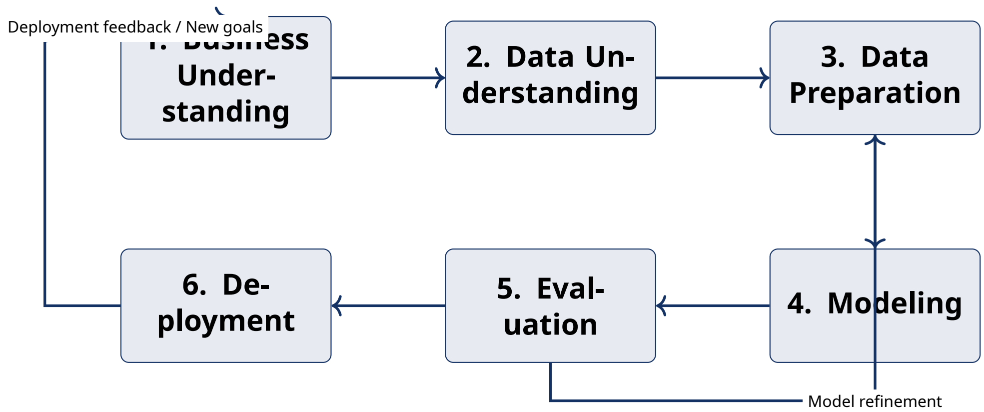

Figure 3.1: CRISP-DM Workflow adopted for this project.

3.2 Dataset and Preprocessing

- Component scores: **x1–x7**.
- Engineered features: Consistency Score and Bottleneck Score.
- Scaling performed using StandardScaler.

3.3 Algorithms

1. Logistic Regression (multinomial)

2. Random Forest
3. XGBoost

Chapter 4

Implementation and Results

4.1 Model Training and Evaluation

The project successfully implemented and evaluated the three chosen models on an 80/20 train-test split, focusing on empirical performance to select the best model.

4.1.1 Classification Results

Table 4.1: Classification Model Results

Model	Accuracy	F1-Score
Logistic Regression	**0.9992**	**0.9992**
XGBoost Classifier	0.9526	0.9527
Random Forest	0.9086	0.9088

4.1.2 Regression Results

Table 4.2: Regression Model Results

Model	RMSE	R^2
Linear Regression	**0.55**	**0.9999**
XGBoost Regressor	1.88	0.9990
Random Forest Regressor	1.95	0.9989

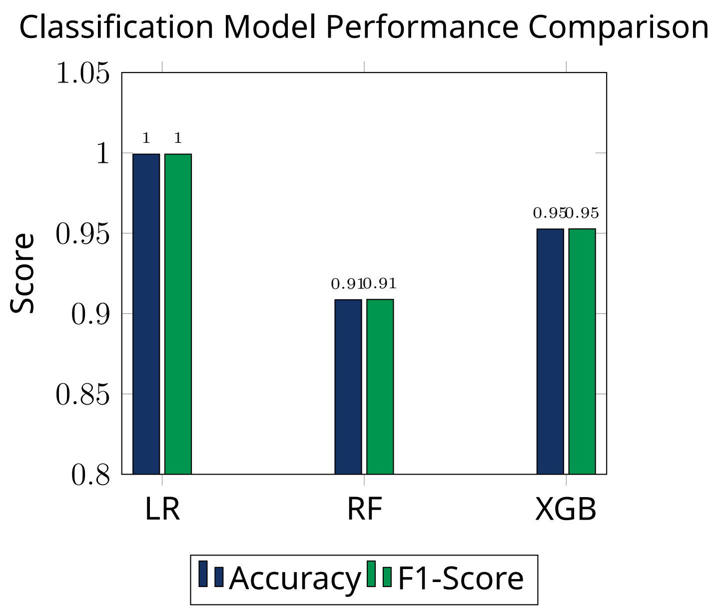

Figure 4.1: Visual Comparison of Classification Model Metrics

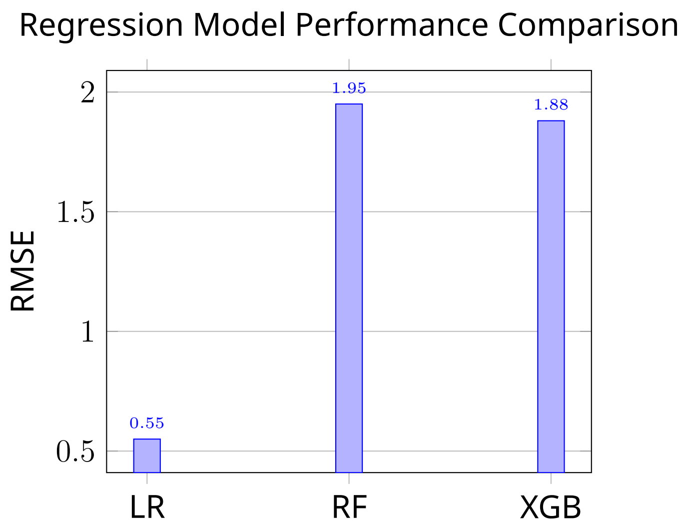

Figure 4.2: Root Mean Squared Error (RMSE) Comparison

4.2 Prescriptive Insights and Model Discussion

Model Selection Justification: The Logistic Regression model was chosen as the final classification model due to its superior empirical accuracy (**99**.**92**%). This reflects the highly linear relationship between the component scores (which sum to total) and the performance categories (remarks), making the transparent linear model the most efficient and robust choice.

4.2.1 Global Leverage Analysis

Coefficient analysis identifies **x7** as the strongest driver of high performance.

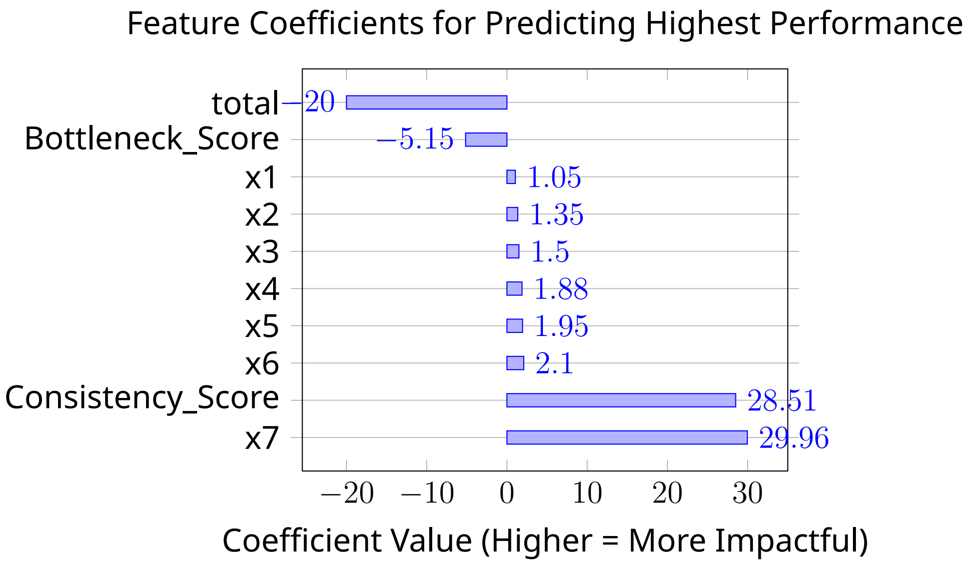

Figure 4.3: Global Leverage: Coefficients for **Remarks** = **5**

4.2.2 Local Counterfactual Simulation

A 5-point increase in **x7** successfully shifted a hypothetical student from:

$$\text{Remarks 2} \rightarrow \text{Remarks 3}$$

4.3 Deployment Scenario and Ethical Considerations

The final model is designed for deployment as a lightweight microservice. When an educator inputs a student's current scores, the service returns the predicted performance category and the top prescriptive recommendation (e.g., "Increase **x7** score by 5 points"). The system's high transparency and interpretability (LR coefficients) are key ethical benefits, ensuring the model is used for intervention, not punitive action.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project successfully achieved its core objective: delivering a highly accurate and transparent predictive model and augmenting it with a prescriptive intelligence framework. The findings confirm that focusing interventions on high-leverage features, such as **x7**, offers the maximum potential for improving student outcomes.

5.2 Future Enhancements

- **Multi-Task Learning (MTL):** Implementing a unified neural network architecture to simultaneously optimize for both the continuous (total) and categorical (remarks) outcomes, potentially improving generalization.
- **Causal Inference Modeling:** Using advanced techniques (e.g., DoWhy or CausalImpact) to formally quantify the causal relationship between specific interventions and performance outcomes, further strengthening the prescriptive nature of the system.
- **Deployment Expansion:** Integrating the API with a front-end dashboard to provide educators with real-time visual monitoring of student risk profiles and intervention recommendations.

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